B450S

Diesel Generator Set，Powered by Baudouin

Main technical data

Rated voltage		380	400	415	440
Prime power	kVA/kW	450 / 360	450 / 360	450 / 360	450 / 360
Standby power	kVA/kW	500 / 400	500 / 400	500 / 400	500 / 400
Ampere	A	684	650	626	591
Frequency	Hz	50			
Rotate speed	RPM	1500			
Phase	P	3			
Power factor	cosφ	0.8			

Structure		Soundproof type
Model		B450S
Tank capacity	L	1650
Dimensions	mm	4422 x 1515 x 2412
Dry weight	kg	4812
Noise Level	dBA@7m	74.7

Load		25%	50%	75%	100%	110%
Fuel consumption	L/h	23.6	44.2	65.6	88.8	110.1

Main Specification

Advantage

- Low fuel consumption
- Optimized system
- High reliability
- High availability
- Long maintenance cycle

Design standards

- Conformite Europeene CE)
- ISO8528-5:2005
- GB/T2820.5-2009

Environmental operating conditions

- Installation place: indoor (well ventilated)
- Ambient temperature: -25℃ to 50℃ (the coolant heater is needed when the temperature is below 5℃)
- Humidity: Less than 90%
- Altitude: Below one thousand (1000) meters.

Performance guarantee

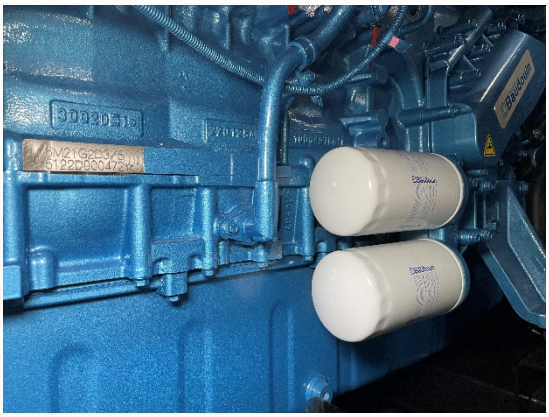
- Product design, manufacturing and performance integrity verified by standards
- Generator set passed transient response test according to ISO8528-5
- Both engine and alternator are prototype and factory tested

Service support

- Provide global product service support

Factory inspection

- Protection devices working test
- Starting ability in normal temperature
- 50% rated power load moment capability
- Voltage deviation and speed variation: 0, 25%, 50%, 75%, 100%, 110%



Power System

Engine

Manufacturer	Baudouin
Model	6M21G550/5
Cylinders and arrangement	6L
Bore: mm	127
Stroke: mm	165
Displacement: L	12.54
Compression ratio	16.0
Rotate speed: RPM	1500
Prime power: kWm	409
Standby power: kWm	450
Rotate speed governor	ECU
Type of injection	Direct

Intake system	Turbocharged
Intake resistance: kPa	≦7.0
Back power: kPa	≦12
Oil capacity: L	34
Coolant capacity: L	45
Battery voltage: V	24
Dimensions: mm	2027×1105×1373
Dry weight: kg	1160

Alternator

Manufacturer	PowerLink
Model	PL5S
Exciter	PMG
AVR model	MX321
Windings	100% copper
Winding pitch	2/3
Number of poles	4
Terminals	12

Insulation class	H
Temperature rising class	H
Drip proof	IP23
Overspeed: RPM	2250
Voltage regulation	±0.5%
Telephone harmonic factor THF	<2%
Telephone interference factor TIF	<50

Control System

Manufacturer	POWERLINK
Model	PLC7420

General functions

- Automatic start/stop control
- Manual/remote start control
- Automatically start when mains is abnormal (AMF)
- Real time monitoring and display of multiple parameters
- RS232, RS485 port and ethernet can be used
- CAN and Modbus communication
- Provide complete control solutions

Monitoring and protection

Oil pressure	Overload
Water temperature	Overcurrent
Rotate speed	Overvoltage
Start	Undervoltage
Running time	Overfrequency
Battery voltage	Underfrequency
……	……

Product Configuration

Standard Configuration

Engine	Alternator	Control switchgear	Canopy (soundproof)	Base frame
Electrical start motor	Insulation class H	PLC control system	Electrogalvanized sheet	Steel base frame
Battery system	Temp. rising class H	GCB, 3P	Anti-corrosion coating	Engine bracket
Speed control system	Drip proof class IP23	Breaker cabinet	Access door	Alternator bracket
Turbocharger	AVR	Communi. connector	Stainless steel hinge	Radiator bracket
Lockable isolator switch		ATS connector	Sound absorbing cotton	Vibration isolators
Battery charger		Mains floating charger		

Fuel system	Lubrication system	Cooling system	Intake/exhaust system	Documents
Base frame fuel tank	Oil pressure sensor	50℃ radiator	Air filter	Installation and operation manual
Fuel level sensor	Oil temp. sensor	Coolant level sensor	Muffler	Test report
Flexible pipe	Oil filter	Jacket water pipe	Exhaust bellows	Wiring diagram
Fuel filter	Manual drain pump	Intercooling pipe	Exhaust pipe and flange	Warranty manual
Fuel inlet	Oil drain ball valve		High temperature protective sleeve	Engine manual
				Standard package

Optional Configuration

Engine	Alternator	Control system	Fuel system	Lubrication system
Jacket water preheater	PMG	GCB, 4P	Fuel-water separator	Electric drain pump
Oil preheater	Anti-condensation heater	ATS cabinet	Fuel three-way valve	
	Treatments against humidity & corrosion	Paralleling control	Daily fuel tank	

Power Class Definition

- Prime Power (PRP): the genset runs continuously with variable load, the number of operating hours is not limited, and 1h overload 10% operation is allowed per 12h, and the average load factor is less than 80% per 24h.
- Standby Power (ESP): operating time does not exceed 500h per year, continuous operating time does not exceed 300h, the average load factor is less than 80% per 24h. Overload operation is not allowed.

Product Statement

- The data of specifications is based on the following standard environmental conditions test
  - Ambient temperature 25˚C
  - Altitude 100m
  - Relative temperature 30%
- Dimensions, weight and other parameters are for reference only, please refer to the final design drawing.

QSV450B6S

DIESEL GENSET SPECIFICATION

450kVA (PRP) / 495kVA (ESP)

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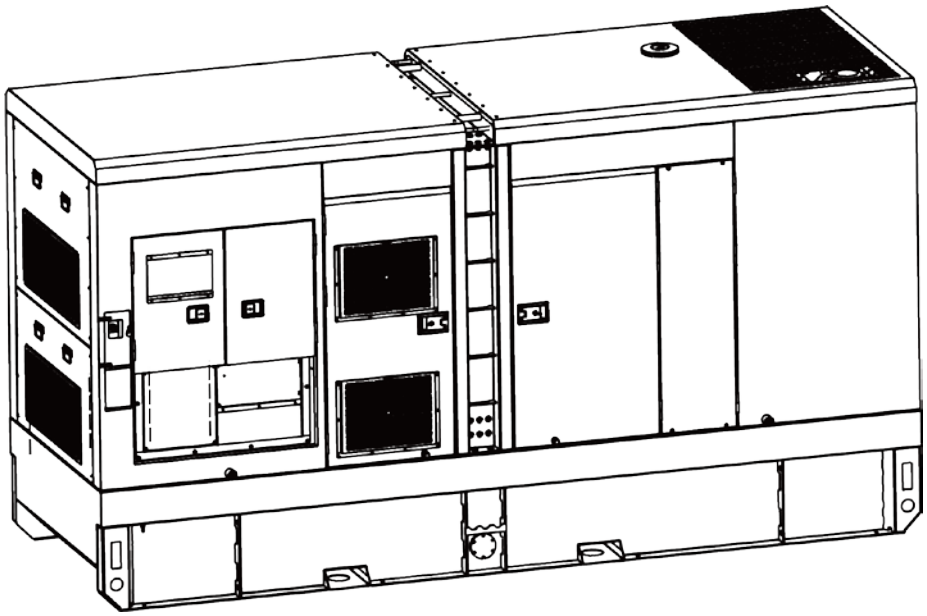


Diesel Genset Specification

QSV450B6S

General technical data

Model	QSV450B6S	Rated voltage V	Current A
Structure	PT	208	1249.112
Prime power (PRP)：kVA / kW	450 / 360	220	1180.978
Standby power (ESP)：kVA / kW	495 / 396	230	1129.631
Frequency: Hz	50	240	1082.564
Rotate speed: RPM	1800		
Phase: P	3		
Power factor: cosφ	0.8	Load	Fuel consumption L/h
Protection class	IP54	25%	N/A
Noise level: dB@7m	76.2	50%	N/A
Tank capacity: L	1650	75%	N/A
Dimensions: mm	4650x1693x2555	100%	N/A
Dry weight: kg	5290	110%	114.3

Main Features

Advantage

- Low fuel consumption
- Optimized system
- High reliability
- High availability

Design Standards

- Conformite Europeenne (CE)
- ISO8528-5:2005

Environmental Operating Conditions

- Installation place: outdoor / indoor (well ventilated).
- Ambient temperature: -25℃ to 45℃ (the coolant heater is needed when the temperature is below 5℃)
- Humidity: Less than 80%
- Altitude: Below one thousand (1000) meters.

Factory Inspection

- Protection devices working test
- Starting ability in normal temperature
- 50% rated power load moment capability
- Load test :0, 25%, 50%, 75%, 100%, 110%

Service Support

- Global product service support

Performance Guarantee

- Product design, manufacturing and performance have been verified by standards
- Generator set has passed transient response test according to ISO8528-5
- Both engine and alternator have passed prototype factory testing.

Diesel Genset Specification

QSV450B6S

Power System

Engine

Manufacturer / brand	Baudouin	Intake system	Turbocharged
Model	6M21G400/6	Intake resistance: kPa	≦3.5
Cylinders	6L	Exhaust back pressure: kPa	≦7.0
Bore: mm	127	Oil capacity: L	30
Stroke: mm	165	Coolant capacity: L	25
Displacement: L	12.54	Battery voltage: V	24
Compression ratio	16.0	Dimensions: mm	2141×1133×1358
Rotate speed: RPM	1800	Dry weight: kg	1260
Prime power: kWm	418		
Standby power: kWm	460		
Rotate speed governor	ECU		
Type of fuel Injection	Direct		

Alternator

Manufacturer / brand	Stamford	Insulation class	H
Model	S4L1D-F4	Temperature rising class	H
Exciter	PMG	Protection class	IP23
Windings	100% copper	Voltage regulation	±1.0%
Winding pitch	2/3	Telephone harmonic factor THF	<2%
Number of poles	4	Telephone interference factor TIF	<50
Terminals	12		

Control System

Brand	DSE
Model	DSE-8610

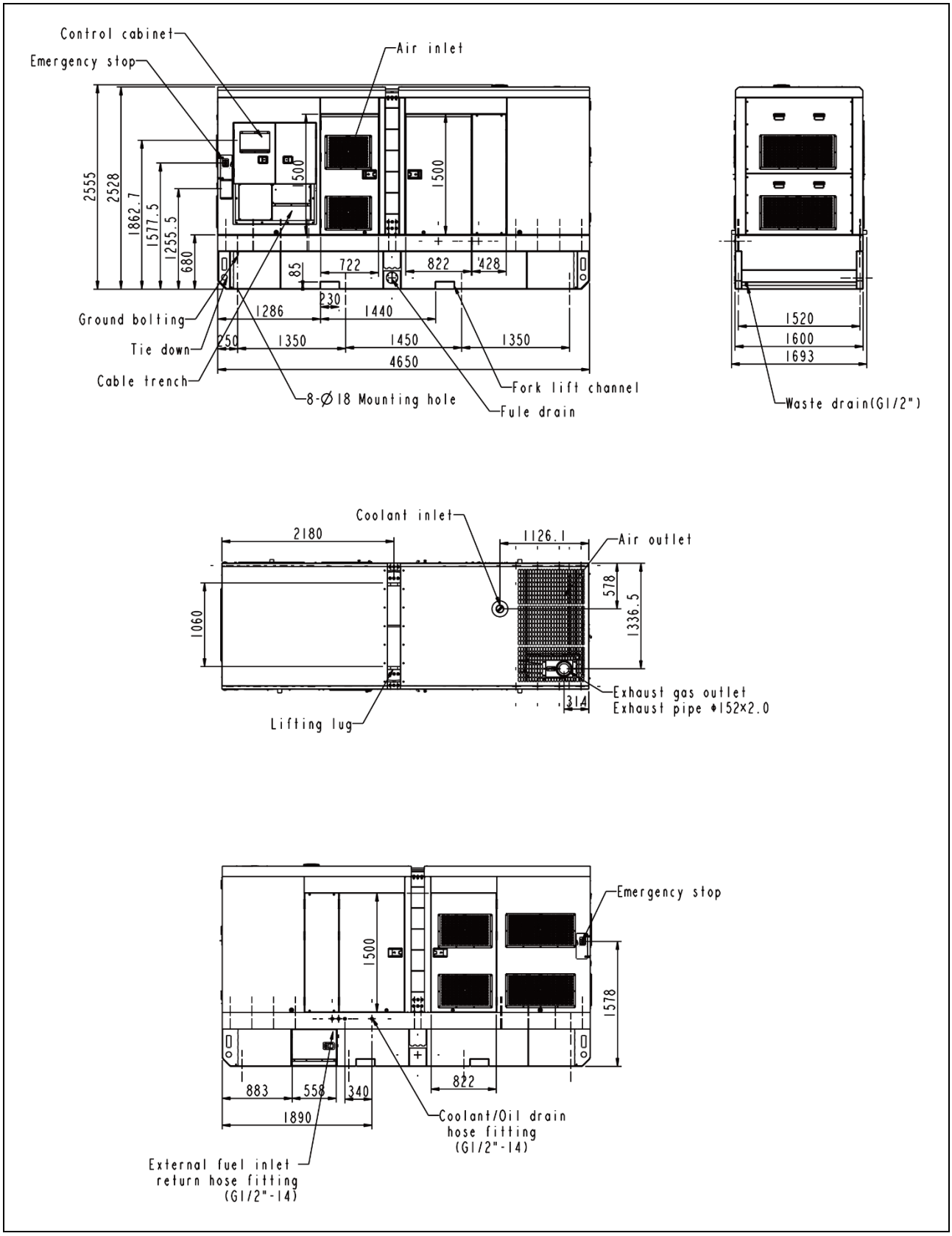
General Functions

- Automatic start / stop control
- Manual/remote start control
- Real time monitoring and display of multiple parameters
- RS232, RS485 interface and Ethernet can be used simultaneously
- CAN and Modbus communication
- Provide complete control solutions

Monitoring and protection

Oil pressure	Over load
Water temperature	Over current
Rotate speed	Over voltage
Start	Under voltage
Running time	Over frequency
Battery voltage	Under frequency
……	……

Product Appearance



Product Configuration

Standard configuration

Engine	Generator	Control switchgear	Canopy (sound-proof)	Base Frame
Electrical start motor	Alternator	PLC control system	SECC Electrogalvanized sheet	Steel base frame
Battery system	H class insulation	Main circuit breaker	Corrosion-resistant coating	Engine bracket
Speed control system	H class temperature rising	Breaker cabinet	Access door	Alternator bracket
Turbocharged	IP23 protection	Communication connector	Stainless steel hinge	Radiator bracket
Lockable isolator switch	AVR voltage regulator	ATS connector	Sound absorbing cotton	Vibration isolators
Battery charger		Mains floating charger		

Fuel system	Lubrication system	Cooling system	Intake / exhaust system	Service documents
Base frame fuel tank	Oil pressure sensor	55℃ radiator	Air filter	Installation and operation manual
Fuel level sensor	Oil temperature sensor	Water level sensor	Exhaust muffler	Test report
Flexible connection pipe	Oil filter	Jacket water pipe	Exhaust bellow	Circuit diagram
Fuel filter	Manual oil drain pump	Intercooling pipe	Exhaust pipe and flange	Warranty manual
	Oil drain ball valve		High temperature protective sleeve	Engine operation and maintenance manual
				Standard package

Optional configuration

Engine	Alternator	Control switchgear	Fuel system	Lubrication system
Jacket water preheater	Anti-condensation heater	4P Circuit breaker	Fuel three-way valve	Electric oil drain pump
Oil preheater	Treatments against humidity and corrosion	ATS cabinet	Daily fuel tank	

Power Class Definition

- Prime Power (PRP): the unit runs continuously with variable load, the number of operating hours is not limited, and 1h overload 10% operation is allowed per 12h, and the average load factor is less than 70% per 24h.
- Standby Power (ESP): operating time does not exceed 500h per year, continuous operating time does not exceed 300h, the average load factor is less than 80% per 24h. Overload operation is not allowed.

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